Lindsay Gross

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AI engineer and product builder at the intersection of technical implementation, trust & safety, and AI governance.

WORK EXPERIENCE

Research Fellow, Deep Tech at Duke, Durham, NC — Feb 2025 – Present

- Authoring a governance case study for Paladin's "AI Tech Stack" initiative, examining security, rogue-agent risks, and accountability in production AI systems.
- Launched the OpenAI-Duke metascience initiative, increasing researcher engagement by ~50% and enabling 5+ new collaborations in Q1.

Trust & Safety Analyst, Tremau, Remote — May 2024 – Sept 2024

- Audited a major social media platform's risk systems, identifying 15+ compliance gaps across online safety, AI accountability, and content moderation; built remediation recommendations adopted by leadership.
- Engaged directly with the client's Trust & Safety Council to incorporate regulatory and civil society insights into their annual risk assessment framework.

AI Ethics, Policy and Research Fellow, Metaphysic.ai, Remote — Feb 2024 – July 2024

- Co-developed a legal and ethical onboarding framework with the Head of Ethics; achieved 100% compliance across 8 client onboardings in 3 months.
- Authored a 20+ page policy paper analyzing US copyright law implications for AI training data misuse.

Cybersecurity Research Assistant, Sanford School of Public Policy, Durham, NC — Dec 2023 – May 2024

- Produced a cybersecurity best practices report on cloud solutions and the NIST framework, referenced by Politico, Forbes, and the World Economic Forum.

SELECTED PROJECTS

SAFE-T (AI Audit Tool) — 2nd place, Duke Society-Centered AI Hackathon

- Built an end-to-end equity audit tool (Python, GeoPandas, Vite, Leaflet.js) that runs 4 bias tests on AI-driven transportation systems; uncovered ~25% undercounting in low-income areas and a 29.5% disparate impact ratio in infrastructure allocation, directly informing Durham safety funding recommendations

Thermal UAV Human Detection

- Developed uncertainty calibration for human detection in adverse weather using PyTorch; focused on making model confidence interpretable and actionable for high-stakes deployment.

Alba (AI Carbon Footprint Extension) — Duke AI Hackathon 2026

- Built a browser extension surfacing real-time energy and water impact estimates for ChatGPT and Gemini queries; designed explainable footprint labels and a prompt optimizer reducing token usage and estimated energy consumption.

AI Audit

- Built an EU AI Act compliance tool (FastAPI, Streamlit, GCP Cloud Run) that classifies AI system risk across Articles 5, 6, 9, 10, and 14 using TF-IDF + logistic regression and a rule-based engine, delivering prioritized remediation plans with MLflow-tracked reproducibility.

EDUCATION

MEng, Artificial Intelligence, Duke University, Durham, NC — May 2025 – May 2026

Coursework: Deep Learning, Reinforcement Learning, Intelligent Agents, AI Product Management, Explainable AI

BA, Public Policy | Digital Intelligence Certificate, Duke University, Durham, NC — Aug 2021 – May 2025

GPA: 3.94 | Cum Laude | Dean's List | VP & Co-founder, Duke Tech for Change (Aug 2024 – Present)

SKILLS

Technical: Python, JavaScript, TypeScript, Node.js, React, PyTorch, NumPy, Pandas, R, Git, APIs, Embeddings, GCP

Product & Research: AI Product Management, Stakeholder Management, Competitive Analysis, Risk Assessment, User Testing, Cross-functional Collaboration

Languages: English (Native), Spanish (Professional Proficiency)